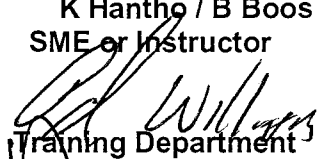
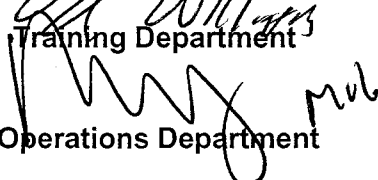


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Administrative		
TASK:	Determine the required actions based on abnormal Secondary Plant chemistry conditions.		
TASK NUMBER:	N1140530402		
JPM NUMBER:	14-01 NRC SRO A1-1		
ALTERNATE PATH:	☒	K/A NUMBER:	2.1.34
IMPORTANCE FACTOR:	3.5		
APPLICABILITY:	EO ☐	RO ☐	STA ☐
		SRO ☒	RO ☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S1.OP-AB.CHEM-0001, Rev. ³²⁴⁻⁹⁶¹⁵ 25 (rev checked 9-12-15)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	10 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	9-12-15
Validated By:	K Hantho / B Boos SME or Instructor	Date:	9-28-15 / 9-30-15
Approved By:	 Training Department	Date:	10-7-15
Approved By:	 Operations Department	Date:	10-9-15
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	SAT	UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:	DATE:		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Administrative

TASK: Determine the required actions based on abnormal Secondary Plant chemistry conditions.

TASK NUMBER: N1140530402

INITIAL CONDITIONS:

While operating at 100% power, Salem Unit 1 has experienced a rise in Sodium levels in 22B main condenser hotwell. Chemistry samples have been reported at 0800 on September 11th:

S/G Blowdown pH: 10.1

S/G Blowdown Chlorides : 38 ppb

S/G Blowdown Sodium: 51 ppb

S/G Blowdown Sulfate: 22 ppb

S/G Blowdown Cation Conductivity: 0.7 uS/cm

All other applicable conditions fall in the "normal" range.

INITIATING CUE:

Using the assumptions that:

- The Secondary Chemistry conditions remain at these levels and cannot be corrected.
- The Condensate Polisher is bypassed.
- SGBD is aligned to the Flashtank at 39k / loop. Chemistry wants 39k / loop to be maintained.
- The SGBD Demineralizer is not the source of the impurities.
- All other systems are in their normal alignment

Determine the actions required over the next 24 hours based on these conditions. Place your response on this JPM tear-off sheet.

Assume that any required plant power changes will only be the minimum required and take the entire length of time allowed.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Determines Secondary Chemistry Action Level 2 is present.
2. Calculates that Unit 1 must be <30% power no later than 2000 on September 11th.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Administrative

TASK: Determine the required actions based on abnormal Secondary Plant chemistry conditions.

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide a clean copy of S1.OP- AB.CHEM-0001, Abnormal Secondary Chemistry, Rev 25.32 <i>25.32A</i>			
			Note: Once entered, the AB would not be exited until the conditions cause were found and isolated.		
	AB.CHEM 3.1	INITIATE Attachment 1, Continuous Action Summary.	Reviews Attachment 1, Continuous Action Summary.		
	3.2	Is Condensate Polishing System in service?	Determines from Initial Conditions that Condensate Polishing System is NOT in service. Goes to Step 3.9.		
	3.9	Is SGBD Demineralizer the source of console alarm?	Determines from Initial Conditions that SGBD Demineralizer is NOT the source of console alarm. Goes to Step 3.11.		
	3.11	NOTIFY Chemistry to initiate sampling IAW SC.CH-AB.ZZ-1104(Q), Response to Abnormal Secondary Chemistry Events.	Notifies Chemistry to initiate sampling IAW SC.CH-AB.ZZ-1104(Q), Response to Abnormal Secondary Chemistry Events.		
	3.12	Is Dissolved Oxygen HI OR Hydrazine LO in alarm?	Determines from Initial Conditions that Dissolved Oxygen HI OR Hydrazine LO is not in alarm.		
	3.13	Is Condensate Polishing System in service?	Determines from Initial Conditions that Condensate Polishing System is not service.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Administrative

TASK: Determine the required actions based on abnormal Secondary Plant chemistry conditions.

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.14	Is Condensate Polishing System Available?	Determines that the Condensate Polishing System is available based on Note preceding step 3.14 that the Normal Condensate Polisher bypass alignment is beds filled, vented, and aligned for service.		
*	3.15	PLACE Condensate Polishing in service as follows:	Determines Condensate Polishing System will be placed in service.		
*	3.16	DETERMINE Chemistry Action Level from Attachment 4, CHEMISTRY ACTION LEVELS.	Using Attachment 4, CHEMISTRY ACTION LEVELS, determines Action Level 2 is met with SGBD Sodium at 51 ppb.		
	3.17	Is Chemistry Action Level 3 required?	Determines Chemistry Action Level 3 is not required? Goes to Step 3.28		
	3.28	Is Chemistry Action Level 2 required?	Determines Chemistry Action Level 2 is required.		
*	3.29	INITIATE immediate actions to reduce power to $\leq 30\%$ and achieve that power within 12 hours of entering Action Level 2 or as quickly as safe plant operation permits (Unit 1 with alloy 600TT SG tubing).	Determines Unit 1 must be $\leq 30\%$ power within 12 hours of entering Action Level 2 (by 2000 on September 11 th .) Note: This 12 hour time requirement is different than the Unit 2 time requirement of 24 hours.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Administrative

TASK: Determine the required actions based on abnormal Secondary Plant chemistry conditions.

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.30	Is entry into Chemistry Action Level 2 due to any of the following? • S/G Blowdown Cation Conductivity • S/G Blowdown Sodium • S/G Blowdown Chloride • S/G Blowdown Sulfate	Determines entry into Chemistry Action Level 2 was due to S/G Blowdown Sodium		
	3.31	Is SGBD aligned to 13 condenser (Flashtank)?	Determines from Initial conditions that SGBD is aligned to 13 condenser (Flashtank)		
	3.31 A	TRANSFER all or some SGBD to 12 condenser as follows:	Determines all or some of SGBD must be transferred to 12 condenser.		
			Note: Steps 33-39 are a "do loop" when attempting to identify and isolate source of impurities.		
			Terminate JPM when operator turns in paperwork.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- AD Kn 1. Task description and number, JPM description and number are identified.
- AD Kn 2. Knowledge and Abilities (K/A) references are included.
- AD Kn 3. Performance location specified. (in-plant, control room, or simulator)
- AD Kn 4. Initial setup conditions are identified.
- AD Kn 5. Initiating and terminating Cues are properly identified.
- AD Kn 6. Task standards identified and verified by SME review.
- AD Kn 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- AD Kn 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 25 Date 7/29/15
- AD Kn 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: Karl Hays

Date: 9/29/15

SME/Instructor: Brian Boos (Brian Boos)

Date: 09/30/2015

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

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